

DDTA - Scheda di lettura compilata

SRC-0046 - BA0-R systems-modeling prior art

REVIEW-ONLY: da approvare prima del caricamento nel repository.

Identita bibliografica e accesso

Source ID	SRC-0046
Titolo	Survey of Candidate Model-Based Systems Engineering (MBSE) Methodologies, Revision B
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1. Research problem and contribution

Provides a cursory survey of leading MBSE methodologies. It explicitly defines methodology as a collection of related processes, methods and tools and distinguishes those from the project environment. The report states that it is a survey, not a comparative assessment.

2. Starting artifacts and generated representations

Methodologies differ substantially. Harmony-SE derives functionality and states/modes and allocates them to physical architecture; OOSEM moves from stakeholder needs/requirements through logical and candidate allocated architectures; OPM integrates object/process/state concepts in one model. The diversity is evidence against assuming one established minimal ontology.

3. Traceability, change, automation and human role

The source discusses repositories, work products and methodology/tool integration but predates modern digital engineering. It is most useful for identifying second-ring sources rather than for current tool maturity or empirical effectiveness.

4. Findings, limitations and relationship with DDTA

The survey shows that structure/behavior/state/allocation recur in different methodologies, but with different conceptual organizations. That supports extracting semantic responsibilities before choosing metaclasses. It also yields bibliography candidates: Wymore, Harmony-SE, OOSEM, Vitech graphical representation relations, JPL State Analysis, Dori OPM and its reflective metamodel.

Evidenza utile per BA0/BA1

Pag.	Ruolo	Sezione	Interpretazione DDTA
0	method	1.1 Purpose / 1.2 Scope	Use the source as a map/bibliography seed, not as evidence that one method is superior.
2	foundation	2.1 PMTE definitions	Environment/context is a separate semantic concern from the modeled system elements.
15	comparison	3.1 Harmony-SE	Behavior, state/mode, structure and allocation recur as distinct responsibilities.
18	comparison	3.2 OOSEM	A neutral representation should support transition between logical and allocated views without equating them.

Pag.	Ruolo	Sezione	Interpretazione DDTA
42	challenge	3.6 OPM	Alternative modeling kernels can be much smaller than SysML while preserving multiple semantic dimensions.
59	bibliography	References	Bibliography mining is the primary value of this source for BA0-R.
64	comparison	Appendix OPM concepts	State and transformation semantics recur even in a very different modeling paradigm.

Bibliography mining / follow-up

- Dori 2002 Object-Process Methodology
- Reinhartz-Berger & Dori 2005 reflective OPM metamodel
- Long 2002 Relationships between Common Graphical Representations in Systems Engineering
- OOSEM/Harmony-SE foundational references
- JPL State Analysis sources

Governance note

Questa scheda è un ausilio di review. Le future citazioni di tesi devono risolversi alla fonte registrata e a una posizione esatta nell'excerpt ledger approvato. Nessun concetto esterno diventa automaticamente un BAE DDTA.